12.18 Memorylessness

WORKSHEET

The question

STEP III 2021, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What is $\mathrm{P}(X > x)$ for $x \geq 0$? What is $\mathrm{P}(X > s + t \mid X > s)$?

STAGE 2 The fractional part

Ask yourself $Z < z$ means X lies in one of the intervals $[n, n + z)$. Add up their probabilities.

STAGE 3 The mean of Z

Ask yourself For a random variable on $[0, 1]$, $\mathrm{E}(Z) = \int_0^1 \mathrm{P}(Z > z)\,\mathrm{d}z$. Use it.

STAGE 4 Independence

Ask yourself Compare with $\mathrm{P}(Y = n)\,\mathrm{P}(z_1 < Z < z_2)$, using parts (i) and (ii).

Try this next

1 **Two bus routes.** (From STEP III, 2002.) Show that the minimum of independent exponentials with parameters λ_1 and λ_2 is exponential with parameter $\lambda_1 + \lambda_2$. Buses on route A and route B have exponential gaps with means 15 and 30 minutes, independently. Having just seen one of each leave, show that the median wait for the next bus is about 7 minutes.

2 **Memoryless discrete.** Show that the geometric distribution is memoryless: $\mathrm{P}(Y \geq m + n \mid Y \geq m) = \mathrm{P}(Y \geq n)$.

3 **The waiting-time paradox.** Buses arrive as a Poisson process with rate λ. Show that the gap containing a fixed time t (for large t) has expected length $\frac{2}{\lambda}$, twice the average gap.

4 **The density of Z.** Find the probability density of Z, and sketch it for $\lambda = 1$ and $\lambda = 5$.

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on the exponential distribution and the Poisson process.

- **Sheldon Ross, *Introduction to Probability Models*** (Academic Press). Exponential distributions, Poisson processes and queues.
- **David Spiegelhalter, *The Art of Uncertainty*** (Pelican). Randomness and waiting in everyday life.

The full guided solution, and what the question leads to at university, is in Section 12.18:
https://beyondstep.keithmcnulty.org/ch12/memoryless.html